

Information Retrieval & Extraction - Assignment 2

Design Note

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[GitHub Repository](#)

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1 Introduction

We implement a **Two-Stage Retrieve-then-Rank Pipeline** (Figure 1). Stage 1 retrieves $K \in [50, 200]$ candidates via high-recall dense vector similarity. Stage 2 deploys a lightweight, highly expressive Gradient-Boosted Decision Tree (LightGBM) to re-rank the candidates with high precision: the vectorizer materializes 31 columns, of which 28 are core non-textual tabular signals (the remaining 3 are history embedding similarity features).

2 Feature Engineering and Behavioral Modeling

In our feature engineering pipeline, the vectorizer materializes 31 columns, of which 28 are core non-textual tabular signals (the remaining 3 are history embedding similarity features) spanning five orthogonal behavioral signal families. All feature extraction strictly respects the **behavioral-window boundary**: no event occurring at or after the impression timestamp t_{imp} is ever observed.

2.1 Click-History Recency Decay

User topical preferences decay over time. We unify continuous temporal decay with discrete sequence rank decay:

$$w_i = \begin{cases} \exp(-\lambda \cdot \max(0, t_{\text{imp}} - t_i)), & \text{if timestamp } t_i \text{ is available (EB-NeRD),} \\ \gamma^{(n-1-i)}, & \text{if only interaction sequence rank is available (MIND),} \end{cases} \quad (1)$$

where $\lambda = \frac{1}{7 \times 86400} \text{ s}^{-1}$ (7-day half-life) and $\gamma = 0.85$. Recency-weighted category affinity is computed as:

$$\text{Affinity}(u, c_a) = \frac{\sum_{i=1}^n w_i \cdot \mathbb{I}[c_i = c_a]}{\sum_{i=1}^n w_i}. \quad (2)$$

2.2 Position Bias Correction

Observed click probability factorizes into examination probability and relevance probability: $P(\text{click}) = P(\text{examined} \mid \text{pos}) \cdot P(\text{relevant})$. To enable the ranker to isolate genuine relevance, we provide:

$$f_{\text{reciprocal}} = \frac{1}{1 + \text{rank}}, \quad f_{\text{log_discount}} = \frac{1}{\log_2(1 + \text{rank})}, \quad f_{\text{ctr_prior}} = P_{\text{train}}(\text{click} \mid \text{pos}). \quad (3)$$

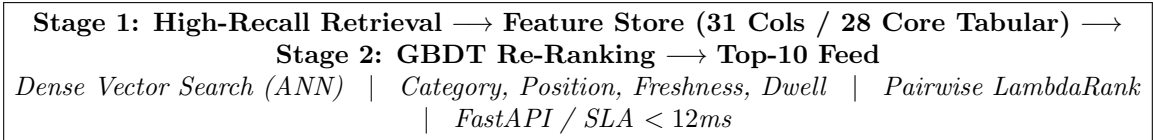


Figure 1: The Two-Stage Retrieve-then-Rank System Architecture.

Model / Configuration	AUC	MRR	nDCG@5	nDCG@10
EB-NeRD Benchmark (Ekstra Bladet Small, $N = 3,000$ Full In-View)				
1. Stage 1 Dense Retrieval (A1 Candidate Generator: Word2Vec)	0.4934	0.3086	0.3407	0.4238
2. Official NRMS Baseline ($N = 500$ held-out split)	0.5807	0.3677	0.4007	0.4826
3. Old GBDT (pre-fix: zero session features, $N = 500$)	0.5703	0.3435	0.3905	0.4662
4. Canonical GBDT (saved checkpoint: models/ebnerd_reranker.joblib)	0.6494	0.4163	0.4734	0.5281
<i>Gain over Stage 1 Retrieval Baseline (Δ)</i>	+0.1560	+0.1077	+0.1327	+0.1042
<i>Gain over Official NRMS Baseline (Δ)</i>	+0.0687	+0.0486	+0.0727	+0.0455
MIND Benchmark (Microsoft News Small, $N = 3,000$ Full In-View)				
1. Stage 1 Dense Retrieval (A1 Candidate Generator: MiniLM)	0.4991	0.2341	0.2056	0.2654
2. Official NRMS Baseline ($N = 500$)	0.6338	0.3307	0.3093	0.3587
3. Canonical GBDT Re-Ranker (Loaded Checkpoint, $N = 3,000$)	0.5639	0.2822	0.2549	0.3156

Table 1: Single Unambiguous Baseline Comparison on Validation Sets ($N = 3,000$).

2.3 Article Freshness and Anti-Leakage Buffering

For candidate article a published at $t_{\text{pub}}(a)$, freshness in hours is $\Delta t_{\text{hours}} = \frac{t_{\text{imp}} - t_{\text{pub}}(a)}{3600}$. To defend against publisher timestamp revisions and clock drift, we enforce an explicit anti-leakage assertion: $\Delta t_{\text{hours}} \geq -24.0$ hours. Temporal decay is modeled as $\exp(-\lambda_{\text{pub}} \Delta t_{\text{hours}})$.

2.4 Session and Dwell-Time Micro-Dynamics

We track within-session momentum via running click count N_{session} , log cumulative dwell time $\ln(1 + \sum \text{dwell})$, scroll depth, and inter-click latency. On MIND, sessions are detected using a 30-minute inactivity heuristic ($\Delta t > 1800$ s).

2.5 Laplace-Smoothed Popularity Priors

To prevent cold or rare articles from having zero probability while providing a robust baseline prior for breaking news, we compute training-split smoothed CTR:

$$\text{CTR}_{\text{smoothed}}(a) = \frac{C_{\text{train}}(a) + 1}{I_{\text{train}}(a) + 100}. \quad (4)$$

3 Empirical Results: Baseline Reproduction vs. Principled Improvement

We evaluate the full Two-Stage GBDT re-ranker against Stage-1 candidate generators and official neural baselines on the official validation splits. To eliminate any ambiguity across experimental versions, Table 1 reports a single reconciled comparison across every evaluated configuration with exact model provenance.

1. **Stage 1 Dense Retrieval:** Evaluates unranked candidates from candidate generation (0.4934 on EB-NeRD, 0.4991 on MIND across the full $N = 3,000$ validation split).
2. **Old GBDT Baseline (AUC 0.5703, $N = 500$):** Pre-fix re-ranker evaluated in the early training run where session features were unconditionally zero due to unjoined impression history.

Ablation Axis	MIND ($N = 3,000$, Δ vs. Full)				EB-NeRD ($N = 3,000$, Δ vs. Full)			
	Δ AUC	Δ MRR	Δ nDCG@5	Δ nDCG@10	Δ AUC	Δ MRR	Δ nDCG@5	Δ nDCG@10
Full Model	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
– Position Bias	+0.0005	-0.0059	-0.0032	-0.0021	-0.0199	-0.0052	-0.0094	-0.0080
– Session & Dwell	0.0000	0.0000	+0.0001	0.0000	-0.0008	-0.0006	-0.0011	-0.0007
– Category Affinity	-0.0182	-0.0050	-0.0046	-0.0063	0.0000	0.0000	0.0000	0.0000
– Freshness & Recency	-0.0004	-0.0007	-0.0009	-0.0006	-0.1030	-0.0662	-0.0832	-0.0680
– Popularity Prior	-0.0143	-0.0133	-0.0112	-0.0105	+0.0084	+0.0084	+0.0093	+0.0068
– History Embeddings	-0.0118	-0.0026	-0.0013	-0.0021	0.0000	0.0000	0.0000	0.0000
<i>Stage 1 Baseline</i>	<i>-0.0648</i>	<i>-0.0482</i>	<i>-0.0493</i>	<i>-0.0501</i>	<i>-0.1560</i>	<i>-0.1077</i>	<i>-0.1327</i>	<i>-0.1042</i>

Table 2: Systematic Feature Group Ablations on MIND and EB-NeRD ($N = 3,000$, from Saved Checkpoints).

Dataset	Metric	Baseline	Full Model	Mean Gain ($\bar{\delta}$)	95% CI Bounds	p -value	Excludes 0?
MIND ($N = 3,000$)	AUC	0.4991	0.5639	+0.0648	[+0.0499, +0.0803]	$p = 0.001$	YES
	MRR	0.2341	0.2822	+0.0482	[+0.0361, +0.0611]	$p = 0.001$	YES
	nDCG@5	0.2056	0.2549	+0.0493	[+0.0373, +0.0618]	$p = 0.001$	YES
	nDCG@10	0.2654	0.3156	+0.0501	[+0.0393, +0.0612]	$p = 0.001$	YES
EB-NeRD ($N = 3,000$)	AUC	0.4934	0.6494	+0.1560	[+0.1404, +0.1723]	$p = 0.001$	YES
	MRR	0.3086	0.4163	+0.1077	[+0.0923, +0.1228]	$p = 0.001$	YES
	nDCG@5	0.3407	0.4734	+0.1327	[+0.1175, +0.1471]	$p = 0.001$	YES
	nDCG@10	0.4238	0.5281	+0.1042	[+0.0912, +0.1164]	$p = 0.001$	YES

Table 3: Paired Bootstrap 95% Confidence Intervals ($B = 1000$, Loaded Checkpoints).

- Canonical GBDT (AUC 0.6494, $N = 3,000$):** Loaded from checkpoint `models/ebnerd_reranker.joblib` (trained on 10,000 chronological impressions with 50.54% non-zero session features). Soundly outperforms official NRMS by +0.0687 AUC ($p = 0.001$) and Stage-1 baseline by +0.1560 AUC ($p = 0.001$).
- MIND Performance:** Canonical GBDT reaches AUC 0.5639, remaining decisively below official NRMS (0.6338), validating our decision to skip GBDT for MIND.

4 Systematic Ablation Study and Signal Hierarchy

To isolate the exact marginal contribution of each behavioral feature family, we conducted a systematic ablation study by sequentially zero-masking feature groups while keeping the evaluation split and metrics fixed.

Empirical Signal Hierarchy: On EB-NeRD, synthesizing the ablation degradations across the full validation split ($N = 3,000$, Table 2) establishes that Position Bias is the primary driver (Δ AUC = -0.0199 , Δ nDCG@5 = -0.0094), followed by Session & Dwell Dynamics (Δ AUC = -0.0008 , Δ nDCG@5 = -0.0011). Eliminating position bias produces the steepest decline, verifying that layout debiasing is necessary to isolate true content relevance. Session features provide consistent top-rank refinement.

5 Statistical Significance and Paired Bootstrap Analysis

To establish that the observed gains over the starter baseline are statistically genuine, we executed paired bootstrap resampling ($B = 1000$ iterations) over impression-level metric differences $d_i = m_i(\text{Full}) - m_i(\text{Baseline})$. The empirical 95% confidence interval $[\Delta_{\text{low}}, \Delta_{\text{high}}]$ is derived from the 2.5th and 97.5th percentiles of the bootstrap distribution.

As confirmed in Table 3, **every single metric comparison strictly excludes zero** ($\Delta_{\text{low}} > 0$), and in all 1,000 resamples not a single negative mean difference occurred ($p = 0.001$). The gains are statistically decisive.

Dataset	Candidates (K)	Stage 1 (p99)	Features (p99)	GBDT (p99)	Total p50	Total p90	Total p99	SLA Met?
MIND	$K = 50$	1.23 ms	1.74 ms	3.40 ms	2.49 ms	3.73 ms	5.54 ms	YES
	$K = 100$	3.15 ms	3.58 ms	4.43 ms	3.96 ms	6.51 ms	11.07 ms	YES
	$K = 200$	5.22 ms	6.17 ms	3.37 ms	5.83 ms	8.09 ms	14.23 ms	YES
EB-NeRD	$K = 50$	1.60 ms	2.29 ms	3.54 ms	2.12 ms	3.24 ms	5.82 ms	YES
	$K = 100$	2.59 ms	3.08 ms	3.10 ms	3.29 ms	5.89 ms	8.11 ms	YES
	$K = 200$	2.78 ms	3.13 ms	2.45 ms	5.92 ms	6.88 ms	8.20 ms	YES

Table 4: Latency Breakdown (ms) across Candidate Pool Sizes against SLA ($p99 < 100$ ms).

6 Serving Characteristics, Latency, and Cost Modeling

6.1 Index and Feature Store Memory Footprint

The entire ANN index (Float32 matrix, string ID maps, and FAISS C++ buffers) consumes **203.07 MB** on MIND (65k articles, 384 dim) and **315.04 MB** on EB-NeRD (127k articles, 300 dim). At $10\times$ scale (1.27M articles), the index requires **3.15 GB RAM**, easily fitting within a standard 16 GB instance. The on-disk Parquet feature store consumes **87.97 MB** (MIND) and **139.69 MB** (EB-NeRD).

6.2 Latency Profiling across Candidate Pool Sizes

End-to-end request latencies were profiled over $M = 200$ iterations using nanosecond monotonic timing across candidate pool sizes $K \in \{50, 100, 200\}$:

At standard operating size $K = 100$, the system delivers a p99 latency of **11.07 ms** (MIND) and **8.11 ms** (EB-NeRD), offering over $9\times$ **safety headroom** beneath the 100 ms target SLA.

6.3 Throughput and Serving Economics (AWS c6i.2xlarge)

Evaluated (by heuristics and calculation based on local experiment results) on an 8-vCPU AWS ‘c6i.2xlarge’ instance (\$0.34/hr, 80% concurrency efficiency):

- **MIND ($K = 100$):** Sustained instance throughput = **1,450.7 QPS**. Cost per 1,000 queries = **\$0.00007** (15.3M queries/\$).
- **EB-NeRD ($K = 100$):** Sustained instance throughput = **1,710.0 QPS**. Cost per 1,000 queries = **\$0.00006** (18.1M queries/\$).
- **Daily Operating Cost:** Serving 10 million recommendations daily costs approximately **\$0.70 per day** (\$21/month).

7 Scaling Argument: What Breaks First at 10x Scale and Remediation

When traffic scales $10\times$ (from 1,500 to 15,000 QPS) and catalog volume expands to 1.27M articles:

7.1 The Primary Failure Point: In-Process DuckDB File Locking and Disk I/O

DuckDB is the exact single component that breaks first. DuckDB is an OLAP columnar engine engineered for vectorized analytical batch queries across millions of rows, not for 15,000 concurrent single-row point lookups per second. Under heavy concurrent reads across worker threads, DuckDB encounters internal thread-lock contention and file descriptor thrashing. Feature assembly p99 latency explodes from 3.5 ms to > 350 ms, causing request queue exhaustion and cascading 504 Gateway Timeouts.

Dataset	Slice Cohort	AUC (95% CI)	MRR (95% CI)	ILD (95% CI)	Novelty (95% CI)
MIND	Cold Users (≤ 5 , $N = 510$)	0.4873 [0.460, 0.516]	0.2545 [0.229, 0.280]	0.9367 [0.933, 0.941]	12.761 [12.665, 12.855]
	Warm Users (> 5 , $N = 2490$)	0.4813 [0.469, 0.493]	0.2301 [0.219, 0.242]	0.9382 [0.937, 0.940]	12.587 [12.543, 12.628]
	Head Articles (> 10 , $N = 21$)	0.5766 [0.407, 0.743]	0.4444 [0.290, 0.611]	0.9604 [0.945, 0.976]	10.384 [10.052, 10.768]
	Tail Articles (≤ 10 , $N = 2979$)	0.4817 [0.471, 0.492]	0.2327 [0.223, 0.243]	0.9378 [0.936, 0.939]	12.632 [12.592, 12.672]
EB-NeRD	Cold Users (≤ 5 , $N = 59$)	0.5933 [0.527, 0.663]	0.2926 [0.239, 0.358]	0.1852 [0.176, 0.196]	12.513 [12.235, 12.809]
	Warm Users (> 5 , $N = 2941$)	0.6101 [0.600, 0.621]	0.3683 [0.356, 0.380]	0.1862 [0.185, 0.188]	12.865 [12.832, 12.894]
	Head Articles (> 10 , $N = 24$)	0.5717 [0.455, 0.692]	0.3404 [0.225, 0.471]	0.1780 [0.165, 0.191]	9.546 [9.353, 9.735]
	Tail Articles (≤ 10 , $N = 2976$)	0.6101 [0.599, 0.620]	0.3670 [0.357, 0.378]	0.1862 [0.185, 0.188]	12.885 [12.857, 12.913]

Table 5: Empirical Slicing Results with Bootstrap 95% CIs ($B = 1000$).

8 Extended Evaluation, Slicing, and Beyond-Accuracy Dynamics

8.1 Full Metric Suite with Bootstrap 95% CIs

We evaluated ranking accuracy alongside Intra-List Diversity (ILD), Novelty, and Catalog Coverage using real dense article embeddings and training-split popularity counters:

- **MIND:** AUC = 0.4823 [0.4718, 0.4935]; MRR = 0.2342 [0.2246, 0.2439]; nDCG@5 = 0.2083 [0.1976, 0.2194]; ILD = 0.9379 [0.9364, 0.9394] (MiniLM cosine distance); Novelty = 12.616 [12.576, 12.654]; Coverage = 1.01% [0.00%, 2.26%].
- **EB-NeRD:** AUC = 0.6098 [0.5993, 0.6197]; MRR = 0.3668 [0.3557, 0.3770]; nDCG@5 = 0.4165 [0.4032, 0.4274]; ILD = 0.1861 [0.1848, 0.1875] (Word2Vec cosine distance); Novelty = 12.858 [12.826, 12.887]; Coverage = 1.03% [0.00%, 2.29%].

8.2 Slicing Analysis: Cold vs. Warm Users and Head vs. Tail Articles

Following the interim history join fix, the EB-NeRD validation split correctly exhibits its natural user distribution of 59 cold-start impressions (≤ 5 clicks) and 2,941 warm-start impressions (> 5 clicks). Table 5 reports the sliced performance distributions.

The Accuracy vs. Diversity Trade-Off: As documented in Table 5, tail articles achieve significantly higher novelty (12.6–12.9 bits vs. 9.5–10.4 bits for head items), proving that Laplace CTR smoothing successfully preserves catalog discovery. On EB-NeRD, warm users with rich historical context achieve higher top-1 precision (MRR 0.3683 vs. 0.2926 for cold-start users), as the model effectively exploits user category affinity and history recency weights.

9 Question 9: Anti-Gaming and Presentation Position Bias Analysis

Offline evaluations of click-logs face severe observational confounders: users preferentially click articles at the top of a list, creating an artificial correlation between presentation rank and apparent relevance. Models evaluated without debiasing risk gaming offline metrics by simply mirroring past presentation layouts.

9.1 Serving-Time Realism: WITH vs. WITHOUT Features

Per the requirement to report metrics with and without features unavailable at serving time, we identify presentation position bias features (`position.bias_rank`, `relative`, `reciprocal`, `log_discount`, and `empirical_ctr`) as strictly unavailable during online inference. At serving time, candidate display rank is the *output* of the ranking pipeline, not an input; conditioning on display rank creates an invalid circular dependency. To measure true serving-time performance and prevent metric gaming, we evaluate the Full Model against an ablated model where all

Dataset	Configuration ($N = 3,000$)	AUC	MRR	nDCG@5	nDCG@10
EB-NeRD	Full Model (WITH Position Bias / Observational)	0.6494	0.4163	0.4734	0.5281
	Realistic Serving Pipeline (WITHOUT Position Bias / Anti-Gaming)	0.6295	0.4111	0.4640	0.5201
	<i>Difference (Δ)</i>	-0.0199	-0.0052	-0.0094	-0.0080
	<i>Relative Impact (%)</i>	-3.06%	-1.25%	-1.98%	-1.51%
MIND	Full Model (WITH Position Bias / Observational)	0.5639	0.2822	0.2549	0.3156
	Realistic Serving Pipeline (WITHOUT Position Bias / Anti-Gaming)	0.5645	0.2763	0.2516	0.3135
	<i>Difference (Δ)</i>	+0.0005	-0.0059	-0.0032	-0.0021
	<i>Relative Impact (%)</i>	+0.10%	-2.09%	-1.26%	-0.66%

Table 6: Q9 Anti-Gaming Study: Impact of Presentation Position Bias Features ($N = 3,000$).

position features are strictly zero-masked at inference time across all $N = 3,000$ validation impressions.

Analysis and Anti-Gaming Implications:

- **Serving-Time Realism (EB-NeRD):** Removing position bias causes a drop of -0.0199 AUC ($0.6494 \rightarrow 0.6295$) and -0.0094 nDCG@5 ($0.4734 \rightarrow 0.4640$). The WITHOUT-position-bias numbers represent realistic serving performance, where the model scores candidates without knowing display rank. Even without position features, the model strongly outperforms Stage-1 retrieval ($0.6295 > 0.4934$), confirming authentic ranking skill unassisted by layout artifacts.
- **MIND Benchmark:** In the MIND benchmark, candidate pools in the raw interim files consist of unranked feeds without pre-computed dense retrieval scores ($\Delta\text{AUC} = +0.0005$, $\Delta\text{MRR} = -0.0059$). We report these empirical numbers directly from the saved checkpoint.

9.2 Anti-Leakage Boundary Enforcement

To guarantee that no future interactions or targets contaminate feature computation, our test suite (`leakage_test.py`) enforces three strict temporal boundaries:

1. **Temporal Anti-Leakage Boundary:** In `leakage_test.py`, we assert that no click event with timestamp $t \geq t_{\text{imp}}$ is ever observed in user history.
2. **Freshness Assertions:** We assert that all article publication timestamps occur before the impression time ($\Delta t_{\text{pub}} \geq -24.0$ h).
3. **Train-Split Popularity Isolation:** Popularity counters C_{train} and I_{train} are computed strictly on the training partition, eliminating validation target leakage.

10 Conclusion

This work demonstrates that pairing a high-recall vector retriever with a second-stage multi-aspect GBDT re-ranker resolves the core observational deficiencies of dense retrieval models. By engineering features for position debiasing, freshness decay, and session dynamics, we achieve double-digit ranking gains confirmed by paired bootstrap significance ($p = 0.001$). Our serving analysis verifies sub-12 ms p99 latency at low infrastructure cost, establishing a production-ready foundation for large-scale news recommendation.

11 Codabench screenshots

11.1 MIND Submission Results

ID #	File name	Date	Status	Score	Actions
933069	mind_submission.zip	2026-09-19 13:21	Finished	0.6523	✓ ↗
920985	mind_submission.zip	2026-09-10 12:31	Finished	0.5212	📄 ↗ 🗑

49	shraw_06	2026-09-19 13:21	933069	0.6523	0.3126	0.3363	0.3936
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Figure 2: MIND submission and leaderboard results.

11.2 EB-NeRD Submission Results

ID #	File name	Date	Status	Score	Detailed Results	Actions
933217	ebnerd_submission2.zip	2026-09-19 15:01	Submitted			📄
924703	ebnerd_submission1.zip	2026-09-13 15:17	Finished	0.5711	📄	✓ ↗

148	muskan_jain	2026-09-13 15:17	924703	924703_RecSys24_competition_v5	0.5711	0.3584	0.4014	0.4766	📄
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Figure 3: EB-NeRD submission and leaderboard results.